

Customer Shopping Behavior Analysis

Uncovering insights into spending patterns, segments, and preferences to guide strategic business decisions.



Project Overview & Data Summary

Project Goal

Analyze 3,900 purchases to understand customer behavior.

- Spending patterns
- Customer segments
- Product preferences
- Subscription behavior

Dataset Snapshot

- Rows: 3,900
- Columns: 18
- Key Features: Demographics, Purchase Details, Shopping Behavior
- Missing Data: 37 values in Review Rating



Exploratory Data Analysis (Python)

1

Data Loading & Exploration

Imported dataset, checked structure and summary statistics.

2

Missing Data Handling

Imputed 'Review Rating' using median of product category.

3

Column Standardization

Renamed columns to snake case for readability.

4

Feature Engineering

Created 'age_group' and 'purchase_frequency_days'.

5

Data Consistency

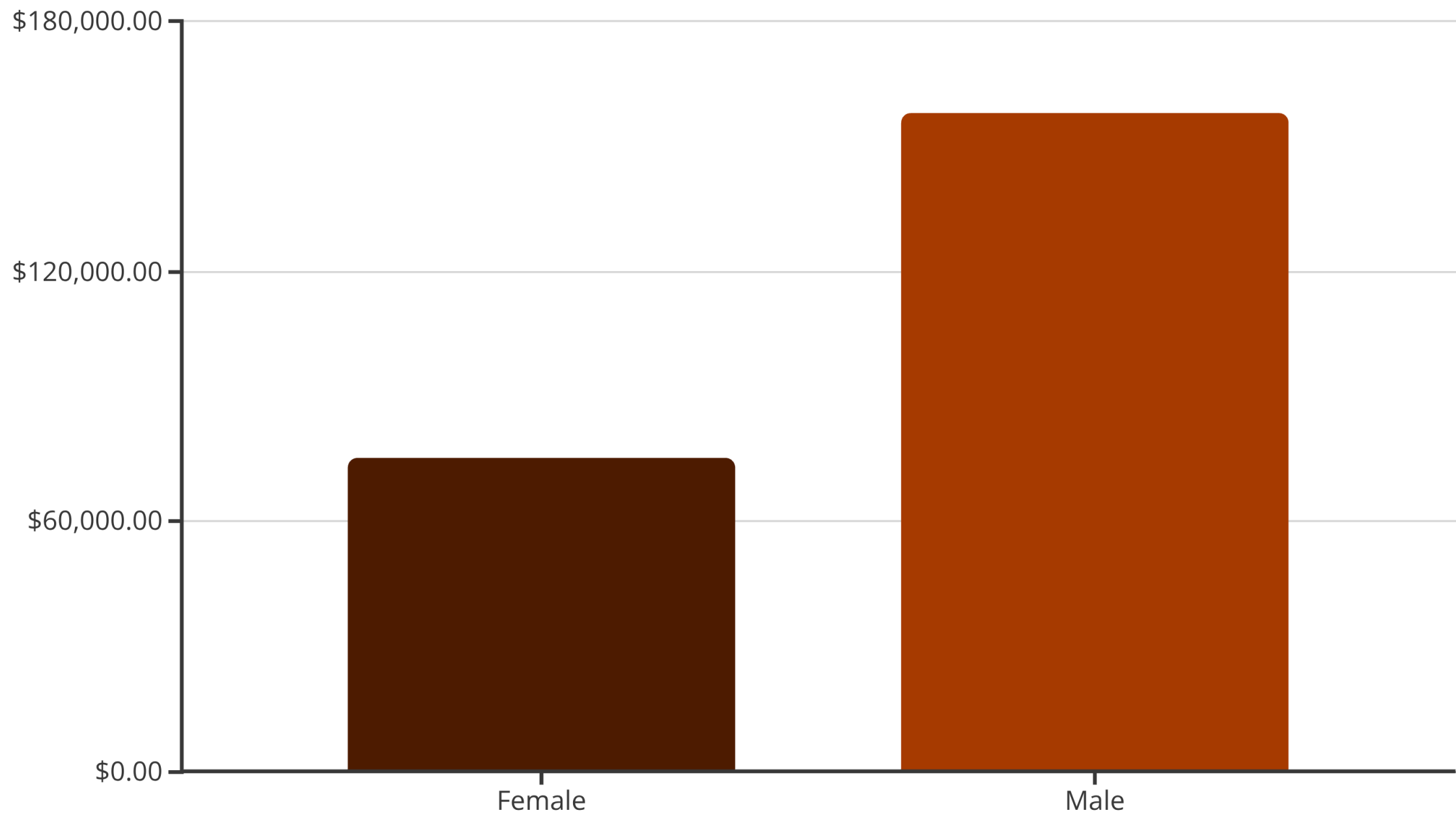
Dropped redundant 'promo_code_used' column.

6

Database Integration

Loaded cleaned data into PostgreSQL for SQL analysis.

Revenue by Gender



Male customers generate significantly more revenue than female customers.





High-Spending Discount Users

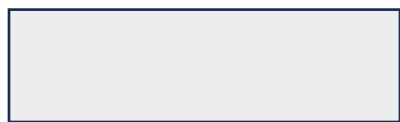
Identified 839 customers who used discounts but still spent above the average purchase amount.

Strategic Discounts

Discounts can attract high-value customers.

Targeted Promotions

Focus on customers who respond to discounts and spend more.



Top 5 Products by Rating & Shipping Comparison

Top 5 Products by Rating

- Gloves: 3.86
- Sandals: 3.84
- Boots: 3.82
- Hat: 3.80
- Skirt: 3.78

Shipping Type Comparison

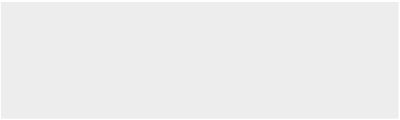
- Standard Shipping: \$58.46 (Avg. Purchase)
- Express Shipping: \$60.48 (Avg. Purchase)



Subscribers vs. Non-Subscribers

Yes	1053	59.49
No	2847	59.87

Non-subscribers have a slightly higher average spend, but subscribers contribute significantly to total revenue.





Discount-Dependent Products & Customer Segmentation

Top 5 Discount-Dependent Products

- Hat: 50.00%
- Sneakers: 49.66%
- Coat: 49.07%
- Sweater: 48.17%
- Pants: 47.37%

Customer Segments

- Loyal: 3116
- Returning: 701
- New: 83

Key Business Recommendations

1

Boost Subscriptions

Promote exclusive benefits for subscribers.

2

Customer Loyalty Programs

Reward repeat buyers to foster loyalty.

3

Review Discount Policy

Balance sales boosts with margin control.

4

Product Positioning

Highlight top-rated and best-selling products.

5

Targeted Marketing

Focus on high-revenue age groups and express-shipping users.

Customer Behavior Dashboard

Interactive Power BI dashboard for visual insights.

- Key Metrics: 3.9K Customers, \$59.76 Avg. Purchase, 3.75 Avg. Review Rating
- Filters: Subscription, Gender, Category, Shipping Type
- Visualizations: Subscription % (Pie), Revenue/Sales by Category (Bar), Revenue/Sales by Age Group (Bar)

